

10

NUMBER AND ALGEBRA

RATIOS AND RATES

Ratios and rates are used in many everyday situations to compare things. Ratios are used when mixing hair colour, mixing paint to the chosen colour, paying rent in a share house and constructing buildings from plans. Rates are used when calculating the amount of paint required for a job, measuring blood alcohol content and preparing a holiday budget.



Chapter outline

		Working mathematically				
10.01	Ratios	U	F			C
10.02	Ratio problems	U	F	PS	R	
10.03	Scale maps and plans	U	F	PS	R	C
10.04	Dividing a quantity in a given ratio	U	F	PS	R	
10.05	Rates	U	F		R	C
10.06	Best buys	U	F	PS	R	C
10.07	Rate problems	U	F	PS	R	C
10.08	Speed	U	F	PS	R	C
10.09	Converting rates	U	F		R	C
10.10	Travel graphs	U	F	PS	R	C

Wordbank

best buy When comparing different brands or sizes during shopping, this is the item with the lowest unit cost and is the best value for money

per (symbol '/') A word used in rates to mean 'for each'

scaled length A length on a map or plan that represents an actual length, usually much smaller, but in proportion to it

speed A rate that compares distance travelled with time taken

travel graph A line graph that describes a journey and shows distance travelled over time

unit price The price of one item or unit, such as 1 mL or 1 g

unitary method A method of finding a quantity by finding one part first

Quiz
Wordbank 10

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communicating

Videos (17):

SkillCheck Why do we have ratios?

10.01 Equivalent ratios • Simplifying ratios 1 • Simplifying ratios 2

10.02 Ratio problems 1 • Ratio problems 2

10.03 Map scales • Scale drawings • 24-hour time

10.04 Dividing a quantity in a given ratio

10.06 Unit pricing

10.07 Rate problems 1 • Rate problems 2

10.08 Speed

10.09 Converting rates 1 • Converting rates 2

10.10 Travel graphs

Twig videos (4):

SkillCheck Ratios: Currency exchange • Ratios: The maths of baking

10.01 The history of the golden ratio

10.08 Speed of the Earth

PhET interactives (3):

10.01 Ratio and proportion • Proportion playground

10.05 Unit rates

Quizzes (5):

- Wordbank 10
- SkillCheck 10
- Mental skills 10
- Language of maths 10
- Test yourself 10

Skillsheets (2):

10.01 Ratios

10.03 24-hour time

Worksheets (16):

10.01 Simplifying ratios • Ratio calculations • Ratios match

10.02 Ratio applications

10.03 Map of Adelaide • Scale drawings

10.05 Rates

10.07 Rate problems • Ratios and rates review

10.08 Speed • What's my speed?

10.09 Converting rates • More rates

10.10 Travel graph stories • The hare and the tortoise

Mind map Ratios and rates

Puzzles (6):

10.01 Simplifying ratios

10.06 Best buys puzzle

10.07 Rate problems

10.08 Speed

10.10 Jane's diary

Language of maths Ratios, rates and time crossword

Technology (1)

10.04 Applying ratios

Presentations (2):

10.04 Shares in ratios

10.08 Rates: Triangle method

Spreadsheet (1):

10.04 Applying ratios

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

- simplify ratios and solve problems involving ratios
- interpret and calculate using scales on maps, plans and images
- divide a quantity in a given ratio
- simplify rates and solve problems involving rates, including speed
- calculate unit prices to determine 'best buys'
- convert between units for rates, for example, kilometres per hour to metres per second
- interpret and draw travel graphs, including the slope of the graph

SkillCheck ANSWERS ON P. 602

1 Copy and complete each conversion.

a 2 m = _____ cm **b** 3 h = _____ min **c** 3000 kg = _____ t

d 2.5 L = _____ mL **e** 380 cm = _____ m **f** 180 mg = _____ g

g 3 min = _____ s **h** 8.5 cm = _____ mm **i** 480 min = _____ h

j 7500 mL = _____ L **k** 9.15 km = _____ m **l** 3840 mm = _____ cm

2 Find the highest common factor (HCF) of each pair of numbers.

a 12 and 18 **b** 35 and 21 **c** 16 and 40

3 Complete each pair of equivalent fractions.

a $\frac{7}{10} = \frac{\square}{30}$ **b** $\frac{3}{5} = \frac{12}{\square}$ **c** $\frac{18}{45} = \frac{\square}{5}$

4 Simplify each fraction.

a $\frac{18}{45}$ **b** $\frac{32}{64}$ **c** $\frac{80}{100}$ **d** $\frac{15}{36}$

5 Evaluate each expression.

a $\frac{2}{5} \times 20$ **b** $\frac{1}{3} \times 15$ **c** $\frac{3}{4} \times 24$ **d** $\frac{5}{6} \times 36$

e 7.6×100 **f** 5.4×10 **g** 0.39×100 **h** 25×10

Quiz
SkillCheck 10

Videos
Why do we have ratios?
Ratios: Currency exchange
Ratios: The maths of baking

10.01

Ratios 10.01

A **ratio** consists of 2 or more numbers that compare the parts or shares of things of the same type, in the same units. For example, if a cake recipe uses sugar to flour in a ratio of 1 to 2, written '1 : 2', it means that for every one part of sugar we need 2 parts of flour.

Each number in a ratio is called a **term** of the ratio.

Equivalent ratios are *equal* ratios and can be found in a similar way to that used for finding **equivalent fractions**.

Equivalent ratios

To find an **equivalent ratio**, multiply or divide each term by the same number.

Skillsheet
Ratios

Puzzle
Simplifying ratios

Worksheets
Simplifying ratios
Ratio calculations
Ratios match

Interactives
Ratio and proportion
Proportion playground

Example 1

Complete each pair of equivalent ratios.

a 6 : 4 = 18 : _____ **b** 3 : 4 = _____ : 24 **c** 12 : 8 = 3 : _____

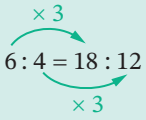
Solution

a To find the missing term, look at the 2 known matching terms, 6 and 18.

6 is *multiplied* by 3 to give 18, so do the same thing to the 4.

$4 \times 3 = 12$

So 6 : 4 = 18 : 12

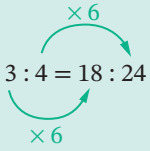


b The 2 known matching terms are 4 and 24.

4 is *multiplied* by 6 to give 24, so do the same thing to the 3.

$3 \times 6 = 18$

So 3 : 4 = 18 : 24

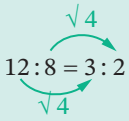


c The 2 known matching terms are 12 and 3.

12 is *divided* by 4 to give 3, so do the same thing to the 8.

$8 \div 4 = 2$

So 12 : 8 = 3 : 2



Videos
Equivalent ratios
The history of the golden ratio

Simplifying ratios

- To simplify a ratio, keep dividing both terms by the same number, preferably a large number such as their highest common factor (HCF), until each term is as small as possible
- If all terms are even, divide by 2 or perhaps 4
- Otherwise, try dividing by the odd numbers 3, 5 or 7

Example 2

Simplify each ratio.

a 25 : 40 **b** 24 : 16 **c** 18 : 30 : 15

Solution

a $25 : 40 = \frac{25}{5} : \frac{40}{5}$

$= 5 : 8$

OR enter 25 : 40 as a fraction $\frac{25}{40}$ on the calculator:

25 $\boxed{a^{b/c}}$ 40 $\boxed{=}$

Divide both terms by their HCF, 5.

$\boxed{a^{b/c}}$ is the fraction key on some calculators.

Videos
Simplifying ratios 1
Simplifying ratios 2

b $24:16 = \frac{24}{8} : \frac{16}{8}$

$= 3:2$

Divide both terms by their HCF, 8.

OR enter 24 : 16 as an improper fraction $\frac{24}{16}$ on the calculator: 24 $\boxed{\square\!\square}$ 16 $\boxed{=}$

Then, to change the mixed numeral answer to an improper fraction, press:

$\boxed{\text{SHIFT}}$ $\boxed{\text{S⇔D}}$ or $\boxed{\text{2nd F}}$ $\boxed{a^b/_c}$

c $18:30:15 = \frac{18}{3} : \frac{30}{3} : \frac{15}{3}$

$= 6:10:5$

Divide all terms by their HCF, 3.

Because there are more than 2 terms in this ratio, the calculator cannot be used here.

Example 3

Simplify each ratio.

a $\frac{3}{5} : \frac{1}{3}$ **b** $0.7 : 0.05$

Solution

If a ratio has terms that are fractions or decimals, it can be simplified by converting the terms to **whole numbers**.

a For fractions, multiply both terms by a common multiple, preferably the **lowest common multiple** (LCM) of the denominators.

$\frac{3}{5} : \frac{1}{3} = \left(\frac{3}{5} \times 15\right) : \left(\frac{1}{3} \times 15\right)$

$= 9:5$

The LCM of 5 and 3 is 15.

b For decimals, multiply both terms by the appropriate power of 10. In this case, multiply by 100 (move the decimal place 2 places to the right).

$0.7:0.05 = (0.7 \times 100) : (0.05 \times 100)$

$= 70:5$

$= 14:1$

Simplifying

Example 4

Simplify the ratio of 2 hours to 1 day.

Solution

First, change the values to the same units.

2 hours : 1 day = 2 hours : 24 hours

1 day = 24 hours

$= 2:24$

$= 1:12$

Video
Simplifying ratios 2

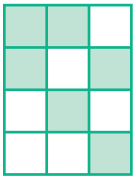
10.01

EXERCISE 10.01 ANSWERS ON P. 602

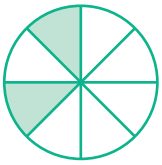
Ratios U F C

1 For each shape, write the ratio of shaded to unshaded parts. C

a



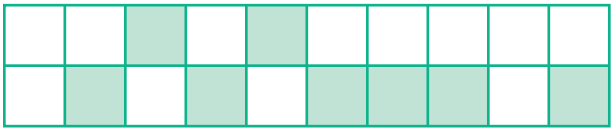
b



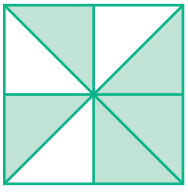
c



d



e



EXAMPLE 1

2 Copy and complete each pair of equivalent ratios.

a	2 : 3 = 8 : _____	**b**	1 : 5 = 2 : _____	**c**	3 : 5 = ____ : 15
d	4 : 7 = _____ : 35	**e**	5 : 8 = 20 : _____	**f**	7 : 12 = 49 : _____
g	5 : 11 = _____ : 66	**h**	3 : 4 = _____ : 100	**i**	2 : 1 = 10 : _____
j	_____ : 9 = 20 : 36	**k**	12 : _____ = 3 : 1	**l**	17 : 34 = _____ : 2
m	_____ : 45 = 6 : 9	**n**	24 : 12 = 4 : _____	**o**	16 : _____ = 2 : 5
p	_____ : 20 = 15 : 60	**q**	24 : 20 = 6 : _____	**r**	50 : 40 = ____ : 20

3 Which of the following ratios is **not** equivalent to the ratio 32 : 48?
Select the correct answer **A**, **B**, **C** or **D**.

A	16 : 24	**B**	4 : 6	**C**	2 : 3	**D**	6 : 8

EXAMPLE 2

4 Simplify each ratio.

a	10 : 100	**b**	12 : 24	**c**	12 : 30	**d**	35 : 49
e	18 : 12	**f**	56 : 24	**g**	1000 : 100	**h**	45 : 99
i	87 : 87	**j**	123 : 321	**k**	51 : 17	**l**	3 : 48
m	8 : 12 : 20	**n**	15 : 20 : 30	**o**	27 : 9 : 36	**p**	14 : 35 : 21 : 49

EXAMPLE 3

5 Simplify each ratio.

a	$\frac{1}{3} : \frac{2}{5}$	**b**	$\frac{1}{4} : \frac{1}{3}$	**c**	$\frac{3}{4} : \frac{2}{3}$	**d**	$\frac{1}{2} : \frac{3}{8}$
e	$\frac{2}{5} : \frac{3}{10}$	**f**	$\frac{4}{5} : \frac{1}{2}$	**g**	$\frac{5}{8} : \frac{1}{4}$	**h**	$\frac{2}{3} : \frac{1}{2}$
i	$\frac{3}{4} : \frac{7}{16}$	**j**	$\frac{4}{5} : \frac{1}{2}$	**k**	$\frac{5}{6} : \frac{2}{5}$	**l**	$\frac{6}{5} : \frac{2}{3}$
m	0.4 : 0.7	**n**	1.3 : 0.8	**o**	0.5 : 0.3	**p**	0.9 : 1.8
q	0.6 : 0.8	**r**	3.6 : 2.4	**s**	0.05 : 0.2	**t**	0.25 : 0.5
u	0.375 : 0.25	**v**	12 : 8.4	**w**	2.4 : 1.2 : 3.6	**x**	4.5 : 1 : 0.9

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EXAMPLE 4

10.01

6 Simplify each ratio.

a	50 cm to 2 m	**b**	300 g to 1.2 kg	**c**	5 days to 7 weeks
d	30 min to 2 hours	**e**	70 cents to $2.10	**f**	2 years to 6 months
g	15 hours to 2 days	**h**	20 mm to 1 m	**i**	4 tonnes to 350 kg
j	25 min to 3 hours	**k**	18 m to 1 km	**l**	8 months to 4 years
m	2 days to 8 hours	**n**	75 cents to $5	**o**	$2.70 : $12

7 On a farm there are 200 orange and mandarin trees in total. If there are 120 orange trees, what is the ratio of orange to mandarin trees? C

8 In a class of 28 students, there are 16 girls. Find, in simplest form, the ratio of: C

a	girls to boys	**b**	boys to girls	**c**	girls to students in the class

9 A store has 30 gas heaters and 20 electric heaters in its warehouse. Find the ratio of: C

a gas heaters to electric heaters

b gas heaters to all heaters

c all heaters to electric heaters

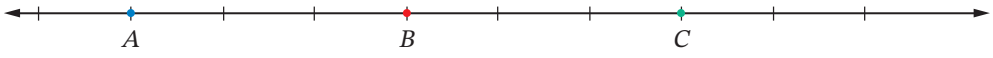
10 A man earns $75 000 a year and spends $63 000 a year. Find the ratio of his savings to earnings. Select **A**, **B**, **C** or **D**. C

A	25 : 21	**B**	21 : 25	**C**	4 : 25	**D**	4: 21

11 Zoe buys goods for $320 and sells them for $380. Find, in simplest form, the ratio of: C

a	the cost price to the selling price	**b**	the selling price to the cost price
c	the profit to the selling price	**d**	the selling price to the profit

12 The line below is divided into units of length as shown.



Find each ratio of lengths. C

a	$AB : BC$	**b**	$AC : AB$	**c**	$BC : AC$	**d**	$AC : BC$

13 A vending machine is filled with bottles and cans. The ratio of the number of bottles to the total number of contents is 3 : 8. C

a What fraction of the contents are cans?

b What is the ratio of the number of cans to the number of bottles?

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14 0.5 m^3 of cement is added to $\frac{3}{8}$ m^3 of metal to make a mixture. What is the ratio of cement to metal? Select **A**, **B**, **C** or **D**. C

A 4 : 3 **B** 7 : 8 **C** 3 : 4 **D** 3 : 16

15 When comparing the rate at which babies are born in different countries (that have different population sizes), we use a base number of 1000. The annual birth rate in Australia is approximately 12 per 1000 people (a ratio of 12 : 1000). The rate in India is approximately 22 per 1000 people, or 22 : 1000. Express each ratio in simplest form. C

10.02 Ratio problems

Worksheet
Ratio applications

Videos
Ratio problems 1
Ratio problems 2

Problems involving ratios can be solved using **equivalent ratios** or the **unitary method**. With the unitary method, we find the size of **one part** first. We have used this method before when calculating with percentages.

Example 5

The ratio of boys to girls in a class is 2 : 3. If there are 10 boys in the class, how many girls are there?

Solution

Method 1: Equivalent ratios

Write the problem as a pair of equivalent ratios.

Boys : girls = 2 : 3 = 10 : ___ (10 boys; ×5 above, ×5 below)

Number of girls = 3 × 5 = 15
There are 15 girls in the class.

Method 2: Unitary method

Boys : girls = 2 : 3
2 parts (boys) = 10
1 part = 10 ÷ 2 = 5
3 parts (girls) = 3 × 5 = 15
There are 15 girls in the class.

'Unitary' means 'one'
Finding one part first.

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Example 6

To make concrete, a builder mixes sand and cement in the ratio 5 : 4. If a mix of concrete contains 20 kg of cement, find:

a the amount of sand in the mix

b the total mass of the mix.



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Solution

a **Method 1: Equivalent ratios**

$\times 5$

Sand : cement = 5 : 4 = __ : 20

$\times 5$

20 kg of cement

Amount of sand = $5 \times 5 = 25$ kg

Method 2: Unitary method

Sand : cement = 5 : 4

4 parts (cement) = 20 kg

1 part = $20 \div 4 = 5$ kg

5 parts (sand) = $5 \times 5 = 25$ kg

b Total mass = 20 kg + 25 kg = 45 kg

Cement and sand

So the total mixture was 45 kg.

EXERCISE 10.02 ANSWERS ON P. 602

Ratio problems U F PS R

1 Alison and Elena buy a length of material and divide it between them in the ratio 2 : 3. If Alison has 5.4 m, what length of material does Elena have? Select the correct answer **A**, **B**, **C** or **D**. PS

A 2.7 m **B** 3.6 m **C** 8.1 m **D** 13.5 m

EXAMPLE 5

2 A tiler uses 4 green tiles to every 3 white ones. How many white tiles are used if 100 green tiles are used? PS

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EXAMPLE 6

3 When making concrete, sand and cement are mixed in the ratio 4 : 1. If 140 kg of cement has been delivered, what mass of sand is needed? PS

4 2 lengths of timber are in the ratio 4 : 7. The longer length is 56 cm. What is the shorter length?

5 In a college, the ratio of teachers to students is 1 : 18. If the college has 80 teachers, how many students are there? PS

6 The speeds of 2 boats are in the ratio 7 : 4. The speed of the slower boat is 10 km/h. Find the speed of the faster boat.

7 In a rectangle, the ratio of the width to the length is 5 : 12. The length is 48 cm. PS R

a Find the width of the rectangle.

b Find the perimeter of the rectangle.

8 An alloy contains copper and iron in the ratio 2 : 5. A quantity of alloy contains 20 kg of copper. What mass of iron does it contain?

9 Toudi, Ash and Felicity share a Lotto prize in the ratio 11 : 8 : 6. Ash received $720. PS R

a How much did Toudi and Felicity each receive?

b What was the total prize money shared?

10 In a triangle, the lengths of the sides are in the ratio 3 : 4 : 5. If the longest side is 45 cm long, find the perimeter of the triangle. Select **A**, **B**, **C** or **D**. PS R

A 57 cm **B** 72 cm **C** 81 cm **D** 108 cm

11 The masses of 2 packets of detergent are in the ratio 3 : 10. PS R

a If the lighter packet has a mass of 1.5 kg, what is the mass of the larger one?

b If the heavier packet costs $12.50 and the lighter packet costs $3.90, which packet is the cheaper per kilogram and by how much?

12 Ed's Farmers' Market buys fruit and vegetables in the ratio 7 : 9. The mass of vegetables ordered is 14.4 tonnes. What is the total mass of produce ordered? PS

13 Samantha and James' heights are in the ratio 7 : 6. If Samantha is 1.75 m tall, how tall is James?

14 In an outback mining town, the ratio of women to men is 2 : 5. If there are 240 women, how many men and women are there in the town altogether? Select **A**, **B**, **C** or **D**. PS R

A 840 **B** 600 **C** 960 **D** 1680

15 To make Superglue, the contents of Tube A and Tube B are mixed in the ratio 3 : 1. PS

a If 21 mL of Tube A is used, how much of Tube B is needed?

b How much glue is made altogether if 21 mL of Tube B is used?

Did you know?

Loose change

All Australian coins are made by mixing copper with other elements.

The $2 and $1 coins were introduced in 1988 to replace notes. They are made of copper, aluminium and nickel in the ratio 46 : 3 : 1.

The 50c, 20c, 10c and 5c coins are made of copper and nickel in the ratio of 3 : 1.

600 grams of aluminium are used to make some $2 coins. How much copper and nickel is used?



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Scale maps and plans 10.03

Scale maps and plans are a special application of ratios in real life. Lengths and distances on scale diagrams are in the same ratio as the real lengths and distances.

Worksheet
Map of Adelaide

Scale ratios

The scale ratio on a scale diagram is written in the form scaled length : real length, where **scaled length** is the length on the diagram.

For example, a scale ratio of 1 : 100 means that the real lengths are 100 times larger than the lengths on the diagram.

Map scales

Map scales are often expressed in the same form as ratios. A scale of 1 cm : 1 km means that 1 cm on the map represents an actual distance of 1 km.

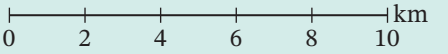
Video
Map scales

Example 7

Simplify each map scale.

a 1 cm : 1 km

b



Solution

a 1 cm : 1 km = 1 cm : 1000 m — 1 km = 1000 m

= 1 cm : 100 000 cm — 1 m = 100 cm

= 1 : 100 000

b The length of the scale from 0 to 10 on the diagram is 5 cm.

So 5 cm on the map represents 10 km of actual distance.

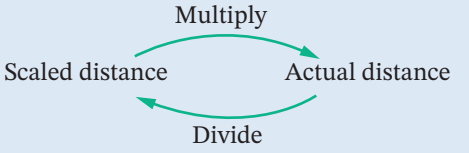
Scale = 5 cm : 10 km

= 5 cm : 10 × 1000 × 100 cm — Convert 10 km to cm.

= 5 cm : 1 000 000 cm

= 1 : 200 000 — Simplifying the ratio.

Map scales



Video
Map scales

Example 8

A map has a scale of 1 : 25 000.

a What is the actual distance if the scaled distance is 4 cm?

b What is the scaled distance if the actual distance is 3.5 km?

Solution

a Scaled distance = 4 cm

Actual distance = 4 × 25 000 cm

= 100 000 cm

= 1000 m — 1 m = 100 cm

= 1 km — 1 km = 1000 m

b Actual distance = 3.5 km

= 3500 m — 1 km = 1000 m

= 350 000 cm — 1 m = 100 cm

Scaled distance = 350 000 cm ÷ 25 000

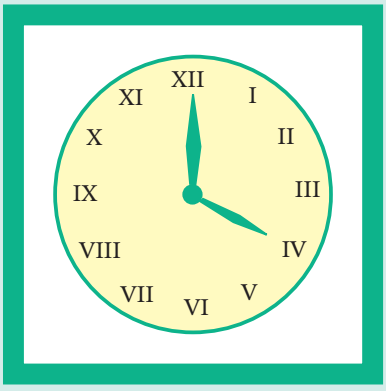
= 14 cm

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Scale drawings

Example 9

This diagram of a clock is drawn to a scale of 1 : 6. Measure its length and calculate its actual length.



Scale 1 : 6

Solution

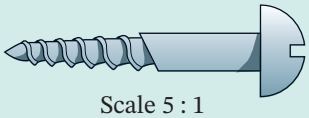
Scaled length = 5 cm by measurement

Actual length = 5 cm × 6

= 30 cm

Example 10

This drawing of a screw is drawn to a scale of 5 : 1. Find its actual length.



Scale 5 : 1

Solution

A scale of 5 : 1 means that the real screw is 5 times **smaller** than the one drawn.

Scaled length = 4 cm by measurement

Actual length = 4 cm ÷ 5

= 0.8 cm

EXERCISE 10.03 ANSWERS ON P. 602

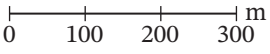
Scale maps and plans U F PS R C

EXAMPLE 7

1 Simplify each map scale. R C

a 1 cm : 5 km **b** 1 mm : 1 m **c** 1 cm : 500 km **d** 1 cm : 25 km

e



f



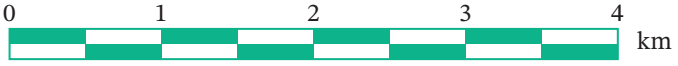
g



h



i



j



EXAMPLE 8

2 A map has a scale of 1 : 50 000. What distance is represented by 64 mm on the map? Select the correct answer **A**, **B**, **C** or **D**. R C

A 0.32 km **B** 3.2 km **C** 32 km **D** 320 km

3 A street map uses a scale of 1 cm : 200 m. PS R C

a Simplify this ratio.

b Find the actual distance, in kilometres, represented by each scaled distance.

i 7 cm **ii** 9.5 cm **iii** 12.4 cm

c Find the scaled distance, in centimetres, used to represent each actual distance.

i 18 km **ii** 1500 m **iii** 9.6 km

4 On a map using a scale of 1 : 10 000 000, the world's longest river, the Nile in Egypt, is the length of an average shoe lace (66.7 cm). How many kilometres long is the River Nile? PS R C

5 The town of Gilgandra is 66 km north of Dubbo. On a map with a scale of 1 : 100 000, what is the scaled distance between the 2 towns? PS R C

6 Lord Howe Island is 2.8 km wide. How long would its scaled width be on a map with a scale of 1 : 50 000? PS R C

7 This map of Nambucca Heads has a scale of 1 : 40 000. PS R C



a Find, in metres, the distance between:

- **i** the Water Towers (C2) and the Catholic Church (F3)
- **ii** the Anglican Church (G4) and the centre of Coronation Park (I4)
- **iii** Rotary Lookout (H6) and Shelley Beach Lookout (J6)
- **iv** the post office (F4) and the Foreshore Caravan Park (D6)

b Find the length of:

- **i** West Street (E4)
- **ii** Piggott Street (D5)

c How long is the lagoon (I7)?

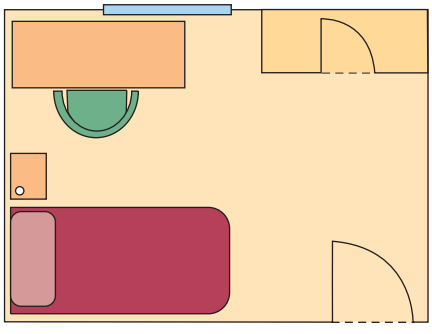
d What are the dimensions of the cemetery (J2)?

e How long is the causeway leading to the Island Golf Course (B7)?

f To train for a fun run, Merridy decides to run 8 km 3 times a week. What distance will this be on the map in centimetres? Outline a possible course for her training run, starting and finishing at the Foreshore Caravan Park (D6).

8 This is a scale plan of a bedroom. R C
By measurement and calculation, find in metres the actual:

a length of the bedroom
b width of the doorway (in cm)
c length of the bed
d length of the window
e length of the table
f area of the bedroom



Scale: 1 : 70

EXAMPLE 9

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9 Measure the length of each scaled-down image below, then use the scale ratio to calculate its actual length in centimetres (metres for the house). R C

a Fish 1 : 3



Shutterstock.com/Elena Blokhina

b House 1 : 300



iStock.com/pamspix

c Pen 1 : 3



Shutterstock.com/Pakhnyushchy

d Tennis racquet 1 : 16



Shutterstock.com/Gearstd

10 This house plan is drawn to a scale of 1 : 140. PS

Measure and calculate to the nearest 0.1 m: R C

a the length of the main bedroom

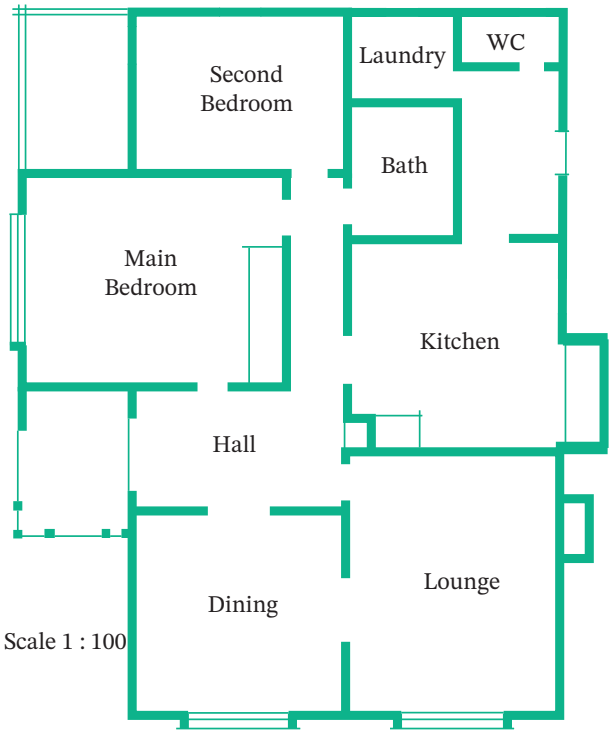
b the length of the window in that room

c the length of the laundry

d the area of the bathroom

e the longer side of the lounge room

f the area of the dining room



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EXAMPLE 10

11 An electronics engineer designs a mobile phone SIM card using a diagram with a scale of 100 : 1. If the scaled drawing is 80 cm long, what is its actual length in millimetres? Select **A**, **B**, **C** or **D**. R C

A 80 **B** 8 **C** 0.8 **D** 0.08

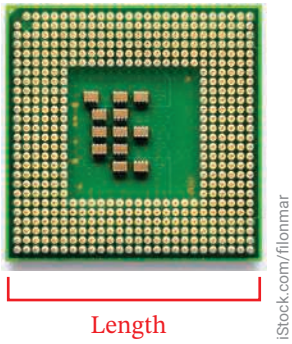
12 Measure the length of each magnified image below, then use the scale ratio to calculate its actual length in millimetres. R C

a Flea 100 : 1



iStock.com/olikim

b Microchip 3 : 1



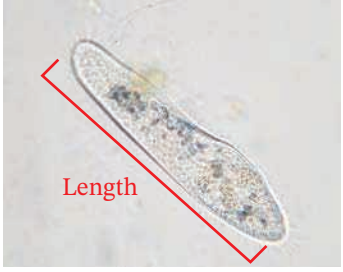
iStock.com/filonmar

c Nut 4 : 3



Shutterstock.com/Alexapicso

d Bacterium 100 : 1



Dreamstime.com/Puntasit Choksawatdikom

Investigation

Scale drawings

1. Measure the dimensions of your bedroom and the dimensions of the items of furniture in it.
2. Using a scale of 1 : 100, draw a scale diagram of your bedroom on graph paper.
3. Draw scale diagrams of each piece of furniture on another piece of graph paper, then cut them out.
4. Rearrange the furniture in your bedroom on the scale diagram to see which arrangement you prefer best.

Quiz
Mental skills 10

Skillsheet
24-hour time

Video
24-hour time

Mental skills 10: Maths without calculators ANSWERS ON P. 603

24-hour time

24-hour time	12-hour time	24-hour time	12-hour time
00:00	12 a.m. (midnight)	12:00	12 p.m. (midday)
01:00	1 a.m.	13:00	1 p.m.
02:00	2 a.m.	14:00	2 p.m.
03:00	3 a.m.	15:00	3 p.m.
04:00	4 a.m.	16:00	4 p.m.
05:00	5 a.m.	17:00	5 p.m.
06:00	6 a.m.	18:00	6 p.m.
07:00	7 a.m.	19:00	7 p.m.
08:00	8 a.m.	20:00	8 p.m.
09:00	9 a.m.	21:00	9 p.m.
10:00	10 a.m.	22:00	10 p.m.
11:00	11 a.m.	23:00	11 p.m.

To convert from **24-hour time** to 12-hour time:

- if it begins with '00', then it is the 12 a.m. (midnight) hour
- if it begins with '12', then it is the 12 p.m. (midday) hour
- if it is less than 12:00, then it is a.m. (morning)
- if it is 13:00 or more, then it is p.m. (afternoon/evening) time: subtract 12 from the hour.

1 Study each example.

a Convert 18:50 to 12-hour time.

18:00 > 13:00, so it is p.m. time, so subtract 12 from the hour.

$18 - 12 = 6$

18:50 = 6:50 p.m.

b Convert 04:30 to 12-hour time.

04:30 < 12:00, so it is a.m. time.

04:30 = 4:30 a.m.

c Convert 00:15 to 12-hour time.

00:15 begins with 00, so it is 12 a.m. time

00:15 = 12:15 a.m.

2 Now convert each time to 12-hour time.

a	08:45	**b**	13:20	**c**	17:50	**d**	00:17
e	21:05	**f**	18:32	**g**	11:15	**h**	02:38
i	14:40	**j**	03:20	**k**	16:55	**l**	23:31
m	01:08	**n**	10:18	**o**	20:00	**p**	06:43

To convert from 12-hour time to 24-hour time:

- if it is the 12 a.m. (midnight) hour, begin with '00'
- if it is 'a.m.' time or the 12 p.m. (midday) hour, write as is but make sure the hour has 2 digits (for example, 02, 09).
- if it is 1 p.m. or later, then add 12 to the hour.

3 Study each example.

a Convert 4:10 a.m. to 24-hour time.

It is 'a.m.' time, so write as a 4-digit number and rename 4 as 04.

4:10 a.m. = 04:10

b Convert 4:10 p.m. to 24-hour time.

It is after 1 p.m., so add 12 to the hour: 4 + 12 = 16

4:10 p.m. = 16:10

c Convert 12:47 a.m. to 24-hour time.

It is in the 12 a.m. (midnight) hour, so change the 12 to 00.

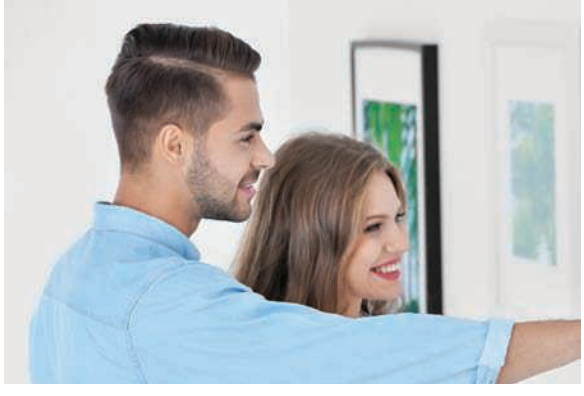
12:47 a.m. = 00:47

4 Now convert each time to 24-hour time.

a	6:35 p.m.	**b**	8:05 a.m.	**c**	11:45 a.m.	**d**	11:20 p.m.
e	2:21 a.m.	**f**	12:30 p.m.	**g**	3:48 p.m.	**h**	7:11 p.m.
i	9:08 a.m.	**j**	9:50 p.m.	**k**	12:42 a.m.	**l**	7:39 a.m.
m	1:59 a.m.	**n**	10:18 p.m.	**o**	10:46 a.m.	**p**	5:23 p.m.

Dividing a quantity in a given ratio 10.04

Elyse and Henry won a cash prize of $420 for winning an art competition, but instead of dividing the money evenly ($210 each), they decide to divide it in the ratio 3 : 4, so that Elyse receives 3 parts while Henry receives 4 parts. Henry receives more because he did more of the work and paid more when buying the materials for the artwork.



Shutterstock.com/Africa Studio

Presentation
Shares in ratios

Technology
Applying ratios

Spreadsheet
Applying ratios

Problems involving dividing a quantity in a given ratio can be solved using the **unitary method** or **fraction method**.

Video
Dividing a quantity in a given ratio

Example 11

Divide a cash prize of $420 between Elyse and Henry in the ratio 3 : 4.

Solution

Method 1: Unitary method

Total number of parts $= 3 + 4 = 7$

7 parts $= \$420$

1 part $= \$420 \div 7$

$= \$60$

Elyse share $(3 \text{ parts}) = 3 \times \$60$

$= \$180$

Henry's share $(4 \text{ parts}) = 4 \times \$60$

$= \$240$

So Elyse receives $180 and Henry receives $240.

Check: $\$180 + \$240 = \$420$

Method 2: Fraction method

Total number of parts $= 3 + 4 = 7$

$$\text{Elyse share } (3 \text{ parts}) = \frac{3}{7} \times \$420$$

$$= \$180$$

$$\text{Henry's share } (4 \text{ parts}) = \frac{4}{7} \times \$420$$

$$= \$240$$

Example 12

Nick, Hong and Rob agreed to divide the profits of their removalist business in the ratio 2 : 3 : 5. If their profit this year was $45 000, find the size of each person's share of the profits.

Solution

Method 1: Unitary method

Total number of parts $= 2 + 3 + 5 = 10$

10 parts $= \$45\,000$

1 part $= \$45\,000 \div 10$

$= \$4500$

Nick's share $= 2 \times \$4500$

$= \$9000$

Hong's share $= 3 \times \$4500$

$= \$13\,500$

Rob's share $= 5 \times \$4500$

$= \$22\,500$

Method 2: Fraction method

Total number of parts $= 2 + 3 + 5 = 10$

$$\text{Nick's share} = \frac{2}{10} \times \$45\,000$$

$$= \$9000$$

$$\text{Hong's share} = \frac{3}{10} \times \$45\,000$$

$$= \$13\,500$$

$$\text{Rob's share} = \frac{5}{10} \times \$45\,000$$

$$= \$22\,500$$

$\therefore$ Nick, Hong and Rob receive $9000, $13 500 and $22 500 respectively.

Check: $\$9000 + \$13\,500 + \$22\,500 = \$45\,000$

EXERCISE 10.04 ANSWERS ON P. 603

Dividing a quantity in a given ratio U F PS R

1 Find the total number of parts if the ratio is:

a 2 : 7 **b** 4 : 1 **c** 3 : 4 **d** 2 : 5 : 6

EXAMPLE 11

2 Divide $500 in the ratio:

a 4 : 1 **b** 7 : 3

3 Phuong, Janelle and Ahmet bought a Lotto ticket for $12. They contributed $3, $3 and $6 respectively to the purchase price. They won $4000 and agree to split the winnings in the same ratio. PS R

a Simplify the ratio 3 : 3 : 6.

b How much prize money does each person get?

4 George and Amal share the weekly rent of $390 in the ratio 7 : 6. What is Amal's share?

5 Divide 450 kg in the ratio:

a 4 : 5 **b** 3 : 2

6 Divide 720 cm in the ratio:

a 1 : 3 : 5 **b** 5 : 3 : 4

EXAMPLE 12

7 Company directors Judy, Gas and Robert share the company profits in the ratio 5 : 3 : 3. Which of the following is the amount that Judy receives in a year when profits are $121 000? Select the correct answer **A**, **B**, **C** or **D**. PS R

A $11 000 **B** $24 000 **C** $33 000 **D** $55 000

8 In Year 8, the ratio of boys to girls is 4 : 5. If there are 225 students in Year 8, find how many girls there are. PS

9 In Year 9, the ratio of boys to girls is 3 : 2. If there are 125 students in Year 9, how many more boys than girls are there? PS R

10 Adam needs to make 800 g of short-crust pastry. Flour and butter are needed in the ratio 3 : 1. How much flour is needed? PS R

11 A company posts 1386 letters in a week. The ratio of local to overseas letters is 2 : 7. How many overseas letters are sent in a week?

12 A truck carries fruit and vegetable boxes in the ratio 5 : 7. If it carries a total mass of 7.5 tonnes, what mass of vegetables does it carry? Select **A**, **B**, **C** or **D**. PS R

A 0.625 t **B** 3.125 t **C** 4.375 t **D** 6.873 t

13 An alloy of mass 176 kg is made from copper and zinc in the ratio 5 : 6. Find the mass of copper in the alloy.

14 At a school, the ratio of students who speak a second language to students who speak only English is 5 : 8. If there are 923 students at the school, how many students speak only English?

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15 A 20 m cable is cut into 3 sections in the ratio 2 : 3 : 5. Find the length of each section.

16 When making mortar, sand and concrete is mixed in the ratio 6 : 1. If we need 280 kg of mortar, how much sand will we need? PS R

17 In a study of 225 000 people, it was found that the ratio of right-handed people to left-handed people was 13 : 2. PS R

a How many left-handed people were there?

b How many more right-handed people than left-handed people were there?

18 Angus earns twice as much as Catriona. If the sum of their wages is $210 000, how much does each earn? R

10.05 Rates

Worksheet
Rates

Interactive
Unit rates

While a ratio compares 2 or more quantities measured in the same units, a **rate** compares 2 quantities measured in **different units**.

A rate shows how one quantity changes with another quantity. We write a rate using a '/' symbol in the form 'something *per* something else'. For example:

- the price of petrol is stated in cents per litre, or c/L
- your heart beats at a rate measured in beats per minute, or beats/min
- the speed of a car is measured in kilometres per hour, or km/h.

The word 'per' means 'for each' so we express a rate '*per* single unit'. For instance, travelling at a rate of 50 km/h means travelling 50 km in *each* hour.

Example 13

Write each statement as a simplified rate.

a A factory produces 87 cars in 3 hours

b Ham costs $50 for 8 kg

Solution

a The production rate would be expressed in cars per hour.

$$\text{Production rate} = \frac{87 \text{ cars}}{3 \text{ hours}}$$
$$= 29 \text{ cars/hour}$$

We can write a rate as a fraction: divide the number of cars by the number of hours

b The cost would be expressed in dollars per kilogram.

$$\text{Cost} = \frac{\$50}{8\text{kg}}$$
$$= \$6.25/\text{kg}$$

Divide the number of dollars by the number of kilograms

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EXERCISE 10.05 ANSWERS ON P. 603

Rates U F R C

1 Write the units suitable for each rate below, in the form _____/_____. R C

a typing speed
b heart rate
c cost of a mobile phone call
d cost of bananas
e a person's wage
f a runner's speed
g population growth
h the cost of water
i population density of a country

2 Write each statement as a simplified rate. C

a 51 sheep in 3 hours
b $10.75 for 2.5 kg
c 208 students for 8 teachers
d 136 points in 4 games
e 546 words in 6 minutes
f 34 articles in 4 hours
g 72 cars in 14 days
h 5040 boxes in 8 hours
i 259 m in 7 seconds
j 46 000 bottles in 50 hours
k 7944 revolutions in 6 minutes
l $175 for 5 hours
m 448 km in 8 hours
n $16.50 for 6 kg
o 114 runs in 24 overs
p 243 km using 30 litres
q $126 for 12 hours
r 2520 kg for 60 hectares

3 The cost of sending a 5.5 kg parcel to Malaysia is $88. What is the postage rate? Select the correct answer **A**, **B**, **C** or **D**. C

A $0.34/kg **B** $16/kg **C** $0.07/kg **D** $484/kg

4 In your own words, explain what is meant by each rate. C

a a speed of 100 km/hour
b a traffic flow of 150 cars/hour
c petrol consumption of 10.3 L/100 km
d a farmer keeping 60 sheep/ha

5 A lift should carry no more than 1600 kg or 20 people. What is this weight allowance, in kg/person? C

6 A pulp mill clears 12 600 hectares of forest in 7 years. At what rate in hectares/year does the mill clear the forest? C

7 The cost of 53 L of petrol is $73.67. Express this cost in c/L. C

8 A complaints hotline took 2190 calls in one year. Calculate the number of calls per month. C

EXAMPLE 13

10.06 Best buys

Video
Unit pricing

When shopping, it is important to compare the prices of different brands or sizes of items and calculate which is the **best buy** ('best value for money'). The biggest container does not always provide the best value. This can be done by comparing the **unit price** (cost of one item or unit) of each brand or size and choosing the cheapest one. A unit could be one gram or one millilitre.

Puzzle
Best buys puzzle

Unit price

Unit price = cost ÷ number of items or units

Supermarkets are required by law to display unit prices to allow you to compare brands or sizes of items. If you look carefully at the price tags on their shelves, you will notice unit prices displayed in small print.

Example 14

Which brand of baked beans is the better buy?



$4.60 $1.65

Solution

Calculating the unit price (cost per gram) for each brand:

Bean There = $4.60 ÷ 500 = $0.0092 / g

Mr Beanz = $1.65 ÷ 200 = $0.00825 / g

Mr Beanz has the lower unit price, so it is the better buy.

EXERCISE 10.06 ANSWERS ON P. 603

Best buys U F PS R C

EXAMPLE 14

1 Find the better buy for each pair of items. PS R C

a A 2 kg box of sultanas for $8.85 or a 1 kg box for $4.65

b 6 notepads for $15.50 or 8 notepads for $20

c 45 g chips for $2.10 or 150 g for $6.50

d 2 tubs of yoghurt for $1.24 or 7 tubs for $4.30

e 3 kg of corn flour for $5.85 or 500 g of corn flour for $1

f 600 mL bottle of fruit juice for $3.20 or a 2.25 L carton for $8.05

2 Michael purchased 400 g of ham for $3.59, while Kim purchased 300 g for $2.54. Who had the better buy? PS R C

3 Which size of Fruit Bix cereal is the best buy? PS R C



$4.75 **$3.49** **$2.51**

4 Happy Tam cat food can be purchased in 4 different packages.

Pack 1: 24 × 250 g cans for $39.99 Pack 2: 12 × 300 g cans for $24.99

Pack 3: 9 × 400 g cans for $25.99 Pack 4: 6 × 1 kg cans for $38.50

Order these packs from best to worst buy. Select the correct answer **A**, **B**, **C** or **D**. PS R C

A 4, 2, 1, 3 **B** 3, 2, 1, 4 **C** 4, 1, 2, 3 **D** 2, 1, 3, 4

5 For each item, find which size is the best value for money. PS R C

a



Alamy Stock Photo/Oliver Hoffmann

Shutterstock.com/Bahadir Yeniceri

b



Shutterstock.com/Marek Tr

Shutterstock.com/Gts

c



Smart Media/Jalna

Smart Media/Jalna

d



Shutterstock.com/Lipowski Milan

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6 For each item, which size gives the best value for money? PS R C

		Small	Medium	Large
a	Washing powder	450 g for $3.99	600 g for $5.15	1 kg for $8.95
b	Margarine	350 g for $4.15	500 g for $5.99	750 g for $8.50
c	Ice cream	500 mL for $1.39	750 mL for $2.30	2 L for $5.99
d	Peas	400 g for $2.05	500 g for $2.49	600 g for $2.50

10.07 Rate problems

Problems involving rates can usually be solved by multiplying or dividing.

Video
Rate problems 1

Worksheets
Rate problems
Ratios and rates review

Puzzle
Rate problems

Solving rate problems

- Write the units of the rate x/y as a fraction: $\frac{x}{y}$
- To find the quantity in the numerator, x, **multiply** by the rate.
- To find the quantity in the denominator, y, **divide** by the rate.

Example 15

Lucas types 55 words per minute. How many words can he type in 20 minutes?

Solution

The units of the rate expressed as a fraction is $\frac{\text{words}}{\text{min}}$.

To find the number of words (the numerator), **multiply** by the rate.

Number of words $= 20 \times 55$
$= 1100$

No. of minutes × words typed per minute

Video
Rate problems 2

Example 16

A factory makes pens at the rate of 50 pens per minute.

a How many hours and minutes will it take to produce 10 000 pens?

b How many pens are produced in an 8-hour day at the factory?

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Solution

The units of the rate expressed as a fraction are $\frac{\text{pens}}{\text{min}}$.

a To find the number of minutes (the denominator), **divide** by the rate.

How many times 50 divides into 10 000.

$$\begin{aligned}\text{Time taken} &= 10\,000 \div 50 \\ &= 200 \text{ minutes} \\ &= \frac{200}{60} \text{ hours} \\ &= 3 \text{ h } 20 \text{ min}\end{aligned}$$

b To find the number of pens (the numerator), **multiply** by the rate.

$$\begin{aligned}\text{No. of pens in one hour} &= 60 \times 50 \\ &= 3000 \\ \text{No. of pens in 8 hours} &= 8 \times 3000 \\ &= 24\,000\end{aligned}$$

1 h = 60 min

No. of mins × pens per min

10.07

EXERCISE 10.07 ANSWERS ON P. 603

Rate problems U F PS R C

EXAMPLE 15

1 Marisol is paid $24.20 per hour. How much will she earn if she works 38 hours in a week?

2 A cricket batsman scores at a rate of 32 runs per innings. How many runs will he score in 5 innings?

EXAMPLE 16

3 A car uses 14 L of petrol to travel 147 km. PS R

a Write this as a rate in km/L.

b How far can the car travel on 20 L of petrol?

4 A farmer can graze 16 sheep per hectare. If he has 21 hectares set aside for sheep, how many sheep can he graze?

5 A tap drips water at a rate of 18 mL/h. How much water would be wasted in one day? PS R

6 An aeroplane can carry a total of 450 passengers per flight. How many flights would it take to carry 2700 passengers? PS R

7 The Wong family needs to buy new carpet for the family home. The carpet costs $125/m and the carpet layers charge $480 to lay it. How much will it cost to carpet the house if the Wongs need 28 metres of carpet? PS R

8 A farmer uses fertiliser at a rate of 28 kg/ha. How many hectares can she cover if she has 200 kg of fertiliser? PS R

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9 Tatiana is paid $17.50/h to babysit. PS R

a How much does she earn for babysitting for 4 hours?

b How long did she babysit to earn $43.75?

10 A plane travelled at 900 km/h. How far did it travel in 20 minutes? PS R

11 **a** Petrol costs $2.03/L. If Rico had $30 in his pocket, how much petrol could he buy (correct to the nearest 0.1 L)?

b If Rico's car can travel 9 km on one litre of petrol, how far can he travel on the petrol he bought with $30? PS R

12 Smokers lose approximately 3% of their lung capacity per year. If Ella with 100% lung capacity starts smoking at 20 years of age, at what age will she have 25% lung capacity? Select the correct answer **A**, **B**, **C** or **D**. PS R

A 75 **B** 28 **C** 45 **D** 33

13 The temperature in Mittagong at 7 a.m. is 8°C. If the temperature rises at a rate of 3°C/h, find: PS R

a the temperature at 10 a.m. and at 2 p.m.

b the time at which the temperature will be 23°C.

14 Keira and Xander take turns mowing a rectangular lawn 60 m long and 40 m wide. PS R

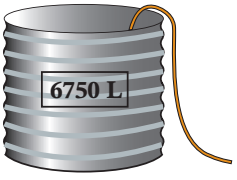
a Find the area of the lawn.

b When Keira cuts the grass, she does it in one hour using an old lawn mower. What is her mowing rate in m^2/min?

c Xander uses a new lawn mower and it takes him 40 minutes. What is his mowing rate in m^2/min?

d How long will it take them to mow the lawn if they mowed together?

15 Jayden is filling this tank with a pipe that pumps water at 1.25 L per second.



How long will it take to fill the tank: PS R C

a in seconds? **b** in minutes? **c** in hours?

16 Mikayla and Jesinta started their holiday on a full tank of petrol.

The reading on the odometer (in km) was

0	3	4	5	6	8

When the tank became empty, the reading was

0	3	5	2	1	8

Petrol cost $1.94/L and they needed to pay $93.12 to fill the tank. PS R

a How many kilometres did they travel on a full tank of petrol?

b How much petrol is there in a full tank?

c How far can their car travel on 1 L of petrol? Answer to the nearest 0.1 km.

Speed

10.08

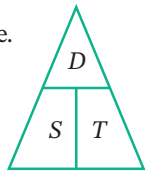
Speed is a rate that compares the distance travelled with the time taken. Average speed is calculated by dividing the distance travelled by the time taken.

Speed

$$\text{Average speed} = \frac{\text{distance travelled}}{\text{time taken}}$$

Speed can be measured in kilometres/hour (km/h) or metres/second (m/s).

The relationship between distance, speed and time can be remembered using this triangle.



If we cover the quantity we are looking for, the rest of the diagram tells us what to do.

- To find the speed, cover the S and we are left with $\frac{D}{T}$, so speed = distance ÷ time
- To find the distance, cover the D and we are left with $S \times T$, so distance = speed × time
- To find the time, cover the T and we are left with $\frac{D}{S}$, so time = distance ÷ speed

Example 17

A bullet train travels 320 km in 1 hour and 20 minutes.

a Calculate its average speed.

b How long will it take to travel 1000 km?

c How far will it travel in 5 hours?

Solution

a $\text{Average speed} = \frac{\text{distance}}{\text{time}}$

$= \frac{320 \text{ km}}{1 \text{ h } 20 \text{ min}}$

$= \frac{320 \text{ km}}{1\frac{1}{3} \text{ h}}$

$= 240 \text{ km/h}$

$20 \text{ min} = \frac{20}{60} \text{ h} = \frac{1}{3} \text{ h}$

b $\text{Time} = \frac{\text{distance}}{\text{speed}}$

$= \frac{1000}{240} \text{ h}$

$= 4\frac{1}{6} \text{ h}$

$= 4 \text{ h } 10 \text{ min}$

By the speed triangle, or because we divide by the rate to find the h in $\frac{\text{km}}{\text{h}}$.

Press $\circ\,'\,''$ or DMS on the calculator.

Worksheets
Speed
What's my speed?

Puzzle
Speed

Presentation
Rates: Triangle method

Videos
Speed

Speed of the Earth

c Distance = speed × time
= 240 × 5
= 1200 km

By the speed triangle, or because we multiply by the rate to find the km in $\frac{\text{km}}{\text{h}}$.

EXERCISE 10.08 ANSWERS ON P. 603

Speed U F PS R C

EXAMPLE 17

1 Find the average speed in km/h for each statement.

a A horse rider travels a distance of 15 km in 3 hours
b A jet travels 2400 km in 5 hours
c A bushwalker walks 25 km in 5 hours
d A cyclist rides 85 km in 5 hours
e A motorbike travels 250 km in 2.5 hours
f A car travels 280 km in 3.5 hours
g A truck travels 390 km in 3 hours and 15 minutes
h A car travels 90 km in 45 minutes
i A motorbike on the race track travels 250 km in 1 hour and 15 minutes
j A boat travels 15 km in 1 hour and 40 minutes
k An athlete runs 200 m in 20 seconds
l A kangaroo travels 1.5 km in 15 minutes

2 Find the average speed in m/s for each statement.

a An athlete runs 200 m in 20 seconds
b A swimmer swims 100 m in 64 seconds
c A bird flies 348 m in 6 seconds
d A cyclist travels 960 m in 2 minutes
e A swimmer swims 1500 m in 16 minutes and 40 seconds

3 Find the distance travelled for each statement.

a A car travels for 8 hours at an average speed of 90 km/h.
b A bushwalker travels for 7 hours and 30 minutes at an average speed of 4 km/h.
c A truck travels for 14 hours at an average speed of 95 km/h.

4 Majid can walk at a speed of 6 km/h. How far can he walk in 30 minutes?

5 A boat travels 24 km in 90 minutes. What is its average speed? Select the correct answer **A**, **B**, **C** or **D**.

A 12 km/h **B** 16 km/h **C** 18 km/h **D** 36 km/h

6 Peter walked to school in 15 minutes, a distance of 1500 metres. Find his speed in: R C

a metres per minute **b** kilometres per hour

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7 Mrs Hassam lives 20 km from the nearest railway station and it takes her 30 minutes to drive there in her car. Find her average speed in km/h.

8 A racing car travels at an average speed of 210 km/h for 2 hours and 20 minutes. How far did it travel? Select **A**, **B**, **C** or **D**. PS R

A 90 km **B** 294 km **C** 462 km **D** 490 km

9 A flywheel rotates at a rate of 2400 revolutions per minute. How many revolutions does it make in 30 seconds? R

10 How long does it take a motor scooter to travel 60 km at a speed of 40 km/h? R

11 There are other units for measuring speed, such as the **knot** and the **mach.** Find what these special terms mean and where they are used. C

12 A bus travels at 75 km/h for 5 hours, and then travels a further distance of 200 km in 4 hours. Find the average speed for the whole journey, correct to one decimal place. PS R

13 A car travels for 3 hours at 60 km/h and then travels a further distance of 100 km at 50 km/h. What is the average speed for the trip? PS R

14 Kieran swims at a speed of 1.85 m/s. What would be his time (to 2 decimal places) for a 100 metre sprint? PS R

15 Nia runs 200 metres in 24 seconds. How far will she run in one minute at the same rate? PS R

Did you know?

Running speeds

The current world records for some of the Olympics running events are shown below, along with the equivalent speed in km/h.

Event	World record	Equivalent speed
Mens		
100 m	9.58 s	37.58 km/h
200 m	19.19 s	37.52 km/h
400 m	43.03 s	33.47 km/h
800 m	1 min 40.91 s	28.54 km/h
1500 m	3 mins 26 s	26.21 km/h
4 × 100 m relay	36.84 s	39.09 km/h
Womens		
100 m	10.49 s	34.52 km/h
200 m	21.34 s	33.74 km/h
400 m	47.6 s	30.25 km/h
800 m	1 min 53.28 s	25.42 km/h
1500 m	3 mins 50.07 s	23.47 km/h
4 × 100 m relay	40.82 s	35.28 km/h



Shutterstock.com/Maxisport

What do you notice about the speeds as the length of the race increases?
Why would the speeds for the 4 × 100 m relays be faster than the speeds for 400 m?

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Technology

Travelling distances and times

Visit a map website such as Google Maps or WhereIs for this investigation to examine travelling distances and times between 2 locations.
Aidan lives near the Wyong Hill Reserve in NSW and is travelling to the Tuggerah shopping centre to see a movie.

1 Use the website to locate both Wyong Hill Reserve and Tuggerah shopping centre, then find the travelling distance and time for Aidan's trip if he is going by car.

2 Find the travelling distance and time for Aidan's trip if he is travelling on foot (walking).

3 How much further would you have to travel if you were going from the Wyong Hill Reserve to the shops by car instead of walking there?

4 Would you recommend that Aidan travel by car or on foot for the trip? Why?

5 **a** Calculate the average speed of the car for the trip.

b Why do you think this average speed is slower than the normal speed limit of 60 km/h?

6 Aidan is meeting Alinta at Tuggerah to see a movie. Alinta lives in Berkeley Vale. Find the travelling distance and time for Alinta's trip if she is travelling by car.

7 If Alinta's actual driving time was 5 minutes, what was her average speed?

8 If Aidan's actual speed was 80 km/h, how many minutes and seconds will his trip take?

9 So will Aidan or Alinta arrive at Tuggerah first? By how many minutes and seconds?

10.09 Converting rates

Worksheets
Converting rates

More rates

What is a speed of 90 km/h in m/s? Sometimes it is useful to be able to convert rates from one set of units to another. Doing this helps us to compare rates for different circumstances.

Converting rates

To convert a rate from one pair of units to another:

- write the first rate as a fraction, including its units
- convert the unit on the numerator of the fraction to the new unit
- convert the unit on the denominator to the new unit
- simplify the rate

Example 18

Convert 90 km/h to m/s.

Solution

$90 \text{ km/h} = \dfrac{90 \text{ km}}{1 \text{ h}}$ — Writing the rate as a fraction.

$90 \text{ km} = 90 \times 1000 \text{ m}$
$= 90\,000 \text{ m}$ — Converting 90 km (the numerator) to metres.

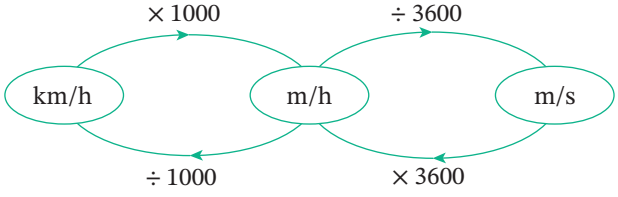
$1 \text{ h} = 60 \times 60 \text{ seconds}$
$= 3600 \text{ s}$ — Converting 1 hour (the denominator) to seconds.

$90 \text{ km/h} = \dfrac{90\,000 \text{ m}}{3600 \text{ s}}$ — Substituting these values back into the rate.

$= 25 \text{ m/s}$ — Dividing to simplify.

A car travelling at a speed of 90 km/h covers 25 metres in one second.

You may find this diagram useful for converting between km/h and m/s.



Example 19

A farmer plants seed at a rate of 45 grams per square metre.

Convert 45 g/m^2 to kg/ha.

Solution

$45 \text{ g/m}^2 = \dfrac{45 \text{ g}}{1 \text{ m}^2}$ — Writing the rate as a fraction.

$45 \text{ g} = 45 \div 1000 \text{ kg}$
$= 0.045 \text{ kg}$ — Converting 45 g to kg using 1 kg = 1000 g.

$1 \text{ m}^2 = 1 \div 10\,000 \text{ ha}$
$= 0.0001 \text{ ha}$ — Converting 1 m^2 to ha using 1 ha = 10 000 m^2.

$45 \text{ g/m}^2 = \dfrac{0.045 \text{ kg}}{0.0001 \text{ ha}}$ — Substituting these values back into the rate.

$= 450 \text{ kg/ha}$ — Dividing to simplify.

A seeding rate of 45 g/m^2 requires 450 kg of seed to cover one hectare.

Videos
Converting rates 1
Converting rates 2

10.09

EXERCISE 10.09 ANSWERS ON P. 604

Converting rates UFRC

In this exercise, round answers to one decimal place if needed. R C

1 Convert:

a $92 424/year to $/month **b** 75 km/h to km/min **c** 5 L/h to L/day

d 3 m/s to m/min **e** 0.8 cm/min to cm/h **f** 78 words/min to words/s

g 8.2 L/100 km to L/km **h** 1750 mL/s to mL/min

EXAMPLE 18

2 Convert each speed into m/s.

a 20 km/h **b** 50 km/h **c** 80 km/h **d** 110 km/h

3 Convert:

a 24c/min to $/hour **b** 8 m/s to km/h **c** 15 mL/s to L/h

d $21.60/h to c/min **e** 60 km/h to m/s **f** 3 t/day to kg/min

4 Asha is walking at a rate of 80 metres per minute. What is her speed in:

a km/h? **b** m/s?

5 Convert each speed to km/h.

a 12.5 m/s **b** 2400 m/min **c** 3500 m/h

EXAMPLE 19

6 Siobhan fertilises her land at the rate of 4 g/m^2. Express this rate in kg/ha.

7 A tap leaks at a rate of 30 mL per minute. What is this rate in litres per hour?

8 The steel mesh used in fencing weighs 1.2 kg/m. What is this rate in:

a g/mm? **b** tonnes/km?

c What do you notice about the answers to **a** and **b**? Can you explain why?

d What is the rate in g/cm?

9 Jesse pumps water from his dam to irrigate his market gardens. The pump delivers water at the rate of 3.5 kL/h. What rate is this expressed as:

a L/h? **b** L/min? **c** L/s?

10 A 5 L container of varnish costs $14.00.

a What is the cost in $/L?

b Express this as a cost in cents/mL.



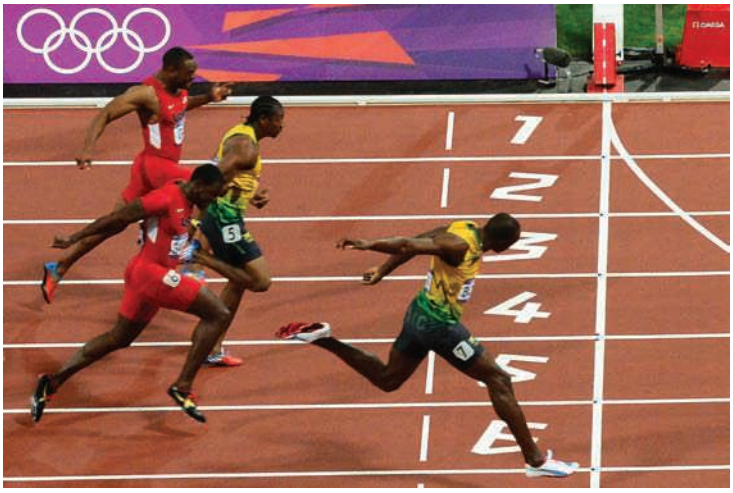
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11 Jamaica's Usain Bolt set the world record for the 100 m sprint in 2009. He ran the distance in 9.58 seconds.

a Express this as a rate in m/s (to 2 decimal places)

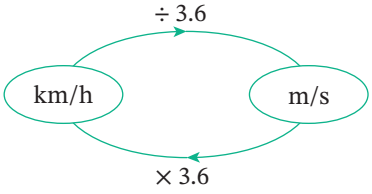
b Convert this speed to km/h (to the nearest km/h).



Getty Images/EMMANUEL DUNAND/AFP

12 The world's fastest cricket bowler is Pakistan's Shoaib Akhtar. In 2003, he bowled a ball 20.12 m in 0.45 seconds. Calculate the speed of the ball he bowled.

13 Explain why the method shown in this diagram works for converting between km/h and m/s. Test it on Questions **2**, **11** and **12**.



10.10 Travel graphs

A **travel graph** or **distance–time graph** is a line graph that describes a journey, by comparing distance (on the vertical axis) with time (on the horizontal axis). The slope or steepness of the graph indicates the speed.

Worksheets
Travel graph stories
The hare and the tortoise

Puzzle
Jane's diary

Video
Travel graphs

Travel graphs

On a **travel graph**:

- a horizontal (flat) section on the graph indicates a stop (the traveller is stationary)
- the steeper the line, the greater the speed (more distance covered in less time)
- a section going down, towards the right, indicates a change in direction or that the traveller is returning towards the start.

Example 20

This graph shows Monique's cycling trip.

a At what time did Monique leave home?

b When was Monique's first stop? How far from home was she?

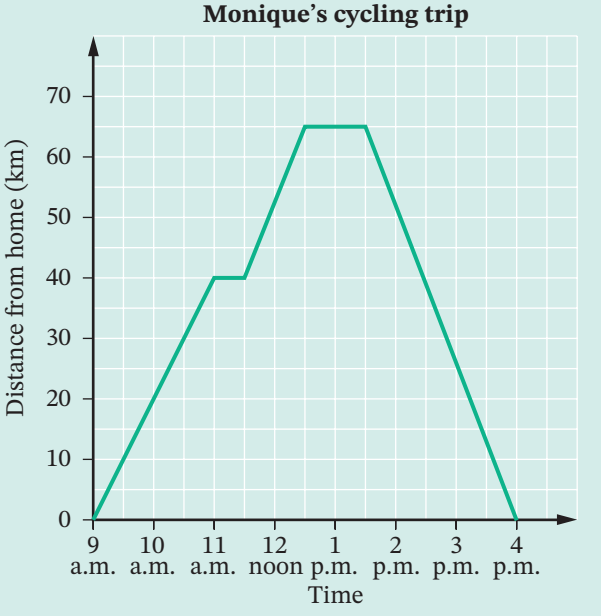
c Find her average speed over the first 2 hours.

d What time was it when Monique began her journey home?

e How far did she travel all together?

f Find her average speed during the trip home.

g For how long did Monique stop altogether during the trip?



Solution

a Monique left home at 9 a.m.

At the start of the graph, when distance = 0.

Where the graph is flat, the distance from home does not change, which means that Monique has stopped

b Monique first stopped at 11 a.m., 40 km from home. This is where the graph is flat.

c Distance = 40 km, time = 2 hours

Average speed $= \frac{40 \text{ km}}{2 \text{ h}} = 20 \text{ km/h}$

d Monique started returning home at 1:30 p.m. This is where the graph points downward.

e Monique travelled 65 km, then returned home.

Total distance $= 2 \times 65 \text{ km} = 130 \text{ km}$

f Distance = 65 km, time $= 2\frac{1}{2}$ hours

Average speed $= \frac{65 \text{ km}}{2\frac{1}{2} \text{ h}} = 26 \text{ km / h}$

g First stop: $\frac{1}{2}$ hour. Second stop: 1 hour

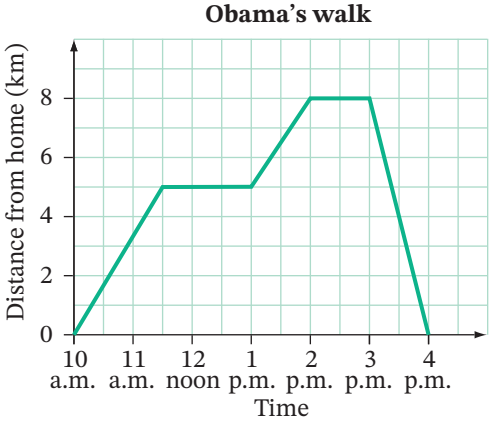
Total stopping time $= \frac{1}{2} + 1 = 1\frac{1}{2}$ hours

EXERCISE 10.10 ANSWERS ON P. 604

Travel graphs U F PS R C

EXAMPLE 20

1 This travel graph shows Obama's return trip to his friend Joe's house. R C



a How long did it take Obama to walk to Joe's house?

b How far is Joe's house from Obama's house?

c Find Obama's average speed:

i before his first stop

ii after his first stop

d How is a higher speed shown on the graph?

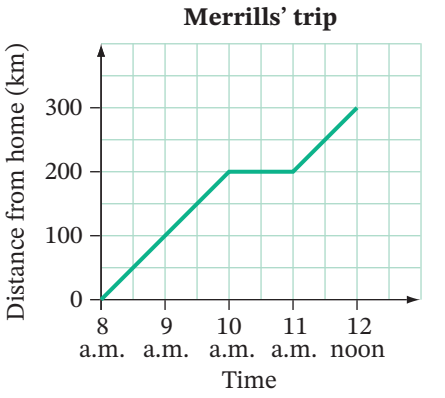
e What was the total distance that Obama travelled?

f Between what times does Obama stop on his walk?

g When did Obama start his trip home?

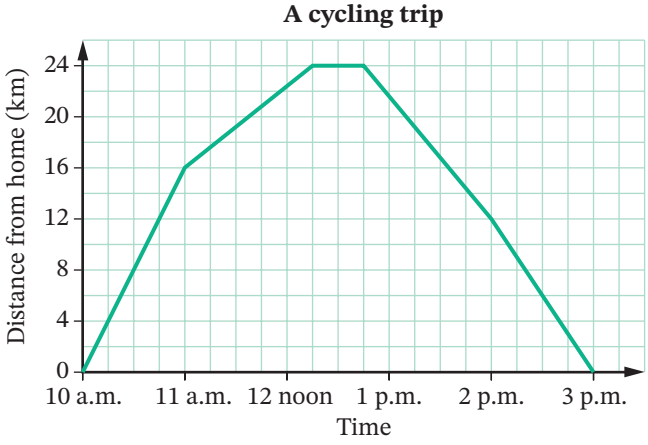
h How long did it take him to walk home?

2 Merrill drives from Sydney to Canberra, stopping to visit friends in Goulburn. R C



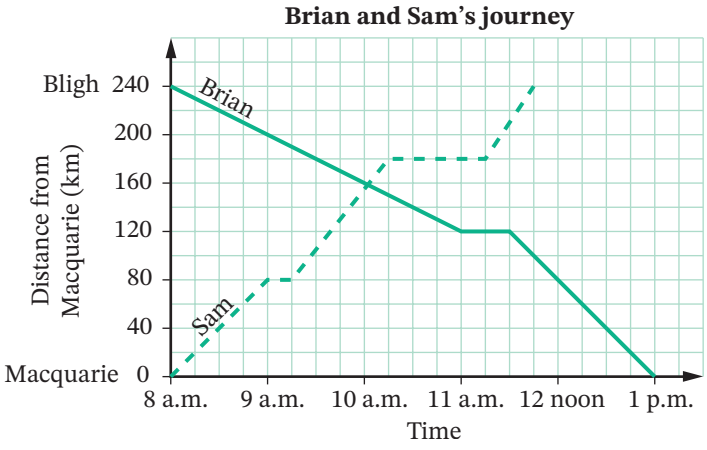
- **a** How far is Canberra from Sydney?
- **b** How long does the trip take?
- **c** For how long does Merrill stop at Goulburn?
- **d** Find her average speed for the journey (excluding stops).
- **e** How far is it from Goulburn to Canberra?
- **f** When does Merrill travel faster: before or after Goulburn?

3 This graph shows a cyclist's day trip. R C



- **a** At what time did the speed of the cyclist
  - **i** increase?
  - **ii** decrease?
- **b** When did the cyclist start to return home?
- **c** How far did the cyclist travel altogether on this day?
- **d** How long did the cyclist spend 'on the road'?
- **e** Find the cyclist's average speed for:
  - **i** the first hour
  - **ii** 11 a.m. to 12:15 p.m.
  - **iii** 12:45 p.m. to 2 p.m.
  - **iv** the entire day

4 Brian travels from Bligh to Macquarie, while Sam travels from Macquarie to Bligh. R C



a How far is it between the 2 towns?

b Who is travelling faster? How can you tell?

c How far is Brian from Macquarie at 11 a.m.?

d How far is Sam from Macquarie at 11 a.m.?

e At what time do Brian and Sam pass each other? How far are they from Macquarie when they pass?

f Is Sam travelling faster before 9 a.m. or after 9:15 a.m.? How does the graph show this?

g Calculate Brian's average speed before he stops.

h For how long did Sam stop altogether on the trip? Select **A**, **B**, **C** or **D**.

A 60 minutes **B** 75 minutes **C** 5 minutes **D** 90 minutes

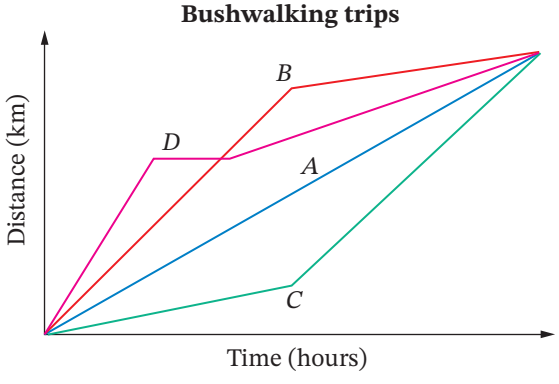
5 Match each bushwalker described below with the correct travel graph. R C

a Santosh maintained the same speed all day.

b Michelle was fast at first, but slowed down.

c Tahlia was the fastest, but then got slower after stopping for lunch.

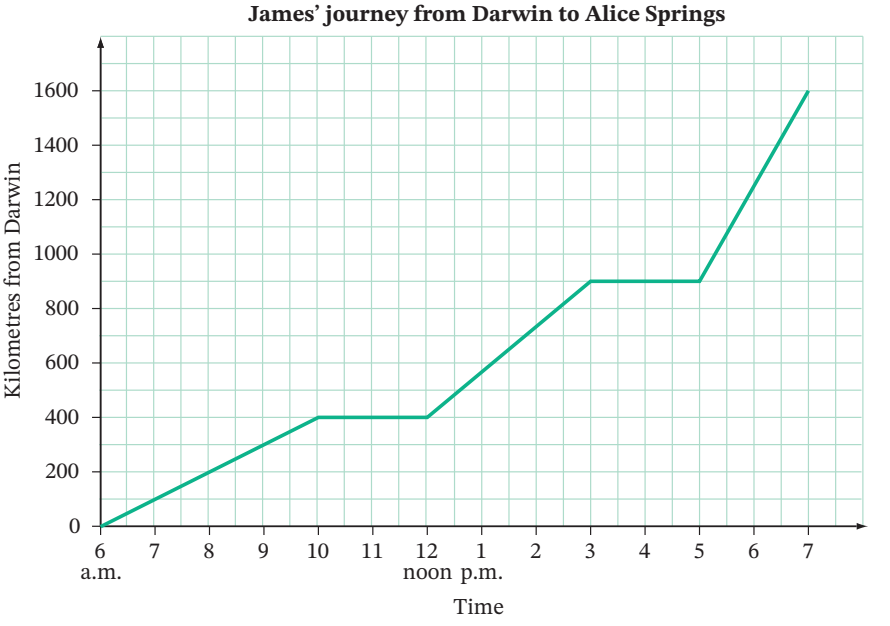
d Jim was slow at first, but picked up speed.



10.10

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6 This graph illustrates James' trip from Darwin to Alice Springs. PS R C



a How far is Alice Springs from Darwin?

b When did James arrive in Alice Springs?

c What was James' average speed in the last 2 hours?

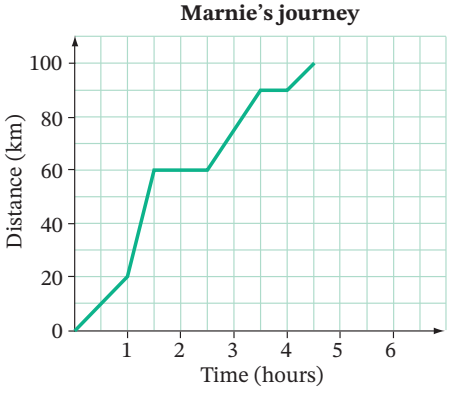
d Which of the following statements is false? Select **A**, **B**, **C** or **D**.

- **A** James stopped for 4 hours altogether.
- **B** James was travelling fastest in the last 2 hours.
- **C** James' speed for the first 4 hours was 100 km/h.
- **D** James' average speed for the whole trip was 120 km/h.

e Samantha leaves Darwin one hour after James and travels at a constant speed of 125 km/h towards Alice Springs. Copy the graph and add Samantha's journey to it.

f Find the approximate time when Samantha overtakes James.

7 This travel graph shows Marnie's cycling journey. Write a story about her ride, based on the information in the graph. C

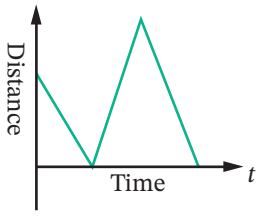


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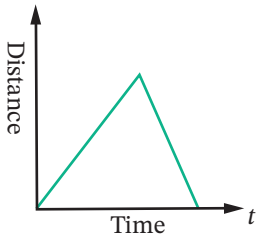
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8 Which one of these graphs could **not** be a travel graph? Why not? R C

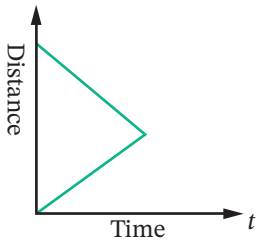
a



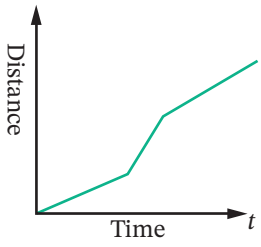
b



c



d



Power plus ANSWERS ON P. 604

1 Australia's annual birth rate in 2021 was approximately 12.1 per 1000. Given that its population then was 26 439 000, approximately how many babies were born in Australia in 2021?

2 Singapore is a small island, but it has a population of 5 780 000, making it the 2nd most densely populated country in the world. Its population density is 8210 persons/km^2.

a What is the area of Singapore, to the nearest square km?

b Australia is the 3rd least-densely populated country in the world, at a rate of only 3.4 persons/km^2 and an area of 7 682 300 km^2. If Australia was as densely populated as Singapore, what would its population be?

3 A spider moves at 1 cm/s. If the spider is in the back left-hand corner of your classroom, find how long (in minutes) it will take to reach:

a the nearest person **b** you **c** the teacher's desk

4 Convert each speed to km/h. Round your answers to 2 decimal places.

a An African cheetah runs at 27 m/s.

b A German peregrine falcon dives at 97 m/s.

c A Tanzanian snake travels at 3.3 m/s.

d A racing cyclist rides at 23 m/s.

5 Work out a formula for converting speed expressed as x m/s to y km/h.

CHAPTER 10 REVIEW

Language of maths

best buy	distance	divide	equivalent
km/h	map	per (/)	plan
rate	ratio	scaled length	scale ratio
simplify	speed	stationary	term
travel graph	unitary method	unit price	

Quiz
Language of maths 10

Puzzle
Ratios, rates and time crossword

1 When comparing the prices and sizes of items to find the **best buy**, do we look for the highest or lowest **unit price**?

2 What type of measurement compares quantities of different types, expressed using 2 units?

3 What is meant by the **scale ratio** of a map or plan?

4 What does an **average speed** of 65 km/h actually mean?

5 How is a change in speed shown on a travel graph?

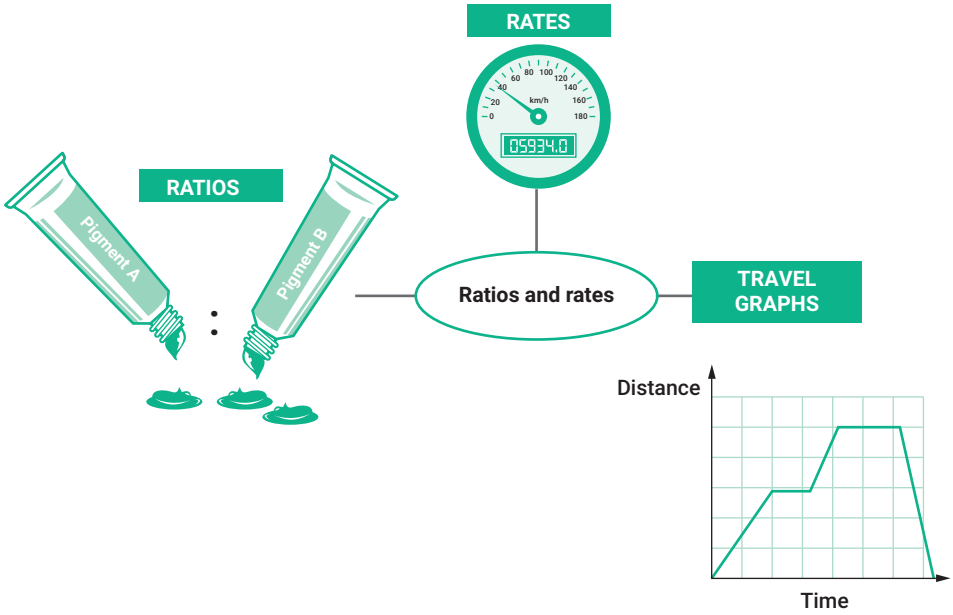
6 What does it mean if a scale drawing has a scale of 20 : 1?

Topic summary

- Give examples of situations where ratios and rates are used.
- What did you learn in this chapter?
- What did you find the most difficult about this topic? Discuss any problems with your teacher or with a friend.

Worksheet
Mind map: Ratios and rates

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



Quiz
Test yourself 10

TEST YOURSELF 10 ANSWERS ON P. 604

10.01

1 Copy and complete each pair of equivalent ratios.

a $1 : 3 = 4 : ____$ **b** $2 : 5 = 6 : ____$ **c** $3 : 7 = ____ : 21$

d $4 : 9 = ____ : 45$ **e** $0.5 : 0.8 = 2 : ____$ **f** $3 : 4 : 5 = ____ : 20 : ____$

10.01

2 Simplify each ratio.

a $12 : 21$ **b** $25 : 75$ **c** $6 : 36$

d $25 : 45$ **e** $18 : 6 : 24$ **f** $5 : 25 : 100$

g $\frac{1}{3} : \frac{1}{2}$ **h** $\frac{3}{4} : \frac{9}{16}$ **i** $1\frac{1}{2} : 2\frac{1}{2}$

j $0.98 : 0.245$ **k** $1.5 : 6$ **l** $91 : 5.6$

m $5 \text{ km} : 200 \text{ m}$ **n** $20 \text{ kg} : 3600 \text{ g}$ **o** $\$25 : \$4.25$

p 18 months: 4 years **q** 10 days: 5 weeks **r** 8 h : 3 days

10.02

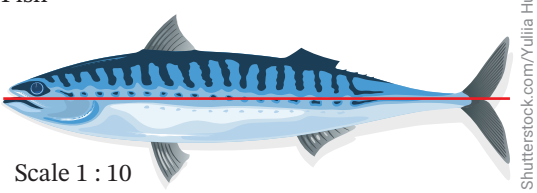
3 **a** The ratio of girls to boys in Year 8 is 4 : 3. If there are 75 boys in Year 8, find how many girls there are.

b Adam and Eve invested in a business in the ratio 5 : 7. If Eve invested \$63 000, how much did Adam invest?

10.03

4 Measure the length of each scale drawing below, and then use the ratio to work out the actual length of the object correct to the nearest centimetre.

a Fish



Scale 1 : 10

Shutterstock.com/Yulia Husar

b Frog



Scale 1 : 4

Shutterstock.com/SGV_Arts

10.03

5 On a tourist map of Sydney, the scale is given by 0 ——— 500 m.

a Write this scale as a simplified ratio.

b Find the actual distance between the following places given the scaled distance.

i Circular Quay station to the Opera House (2.5 cm)

ii Pyrmont Bridge to Parliament House (4.4 cm)

c Find the scaled distance between the following places given the actual distance.

i Art Gallery of NSW to Sydney Tower (875 m)

ii Circular Quay station to Central station (2.5 km)

10.04

6 Ellen and Portia share the weekly rent of \$420 on their apartment in the ratio 4 : 3. How much does Ellen pay? Select the correct answer **A**, **B**, **C** or **D**.

A \$105 **B** \$240 **C** \$180 **D** \$210

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7 Write each statement as a rate in simplified form. 10.05

a $10.50 for 3 kg
b 220 km in 2 hours
c $56.40 for 4 hours
d 260 runs in 50 overs

8 For each pair, determine which is the better buy. 10.06

a A 1 kg box of biscuits for $5.60 or a 700 g carton for $4.00.
b A 350 g can of soup for $4.20 or a 550 g can for $7.15.
c A 1.5 L bottle of lemonade for $2.45 or a 375 mL can for $0.70.
d 3 × 40 g chocolate bars for $2.36 or a 500 g chocolate bar for $9.

9 **a** Mince is $9.99/kg. How much does 5 kg of mince cost? 10.07
b Tiwa earns $31.70/h. How much is she paid for 38 hours work?
c How many litres of petrol can you buy with $60 if petrol costs $1.95/L? Answer to the nearest 0.1 L.
d Fertiliser is used at 20 kg/ha. How many hectares can be covered with 144 kg of fertiliser?

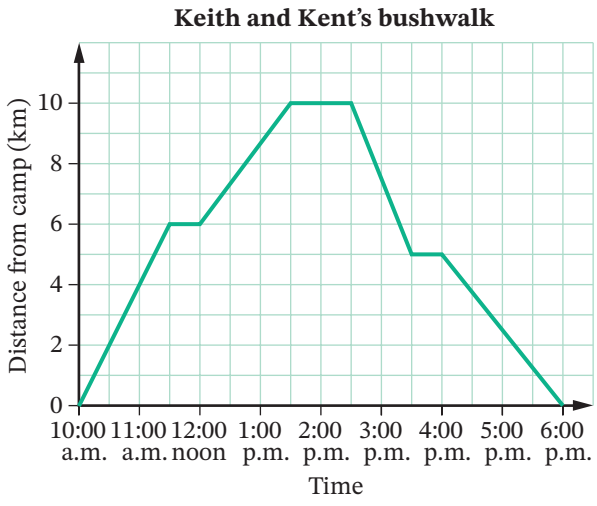
10 Find the speed, in km/h, of a cyclist who travels a distance of 45 km in 2 hours and 15 minutes. 10.08

11 Find the distance covered by a truck that travels for 5 hours and 30 minutes at an average speed of 45 km/h. 10.08

12 **a** Convert a speed of 75 km/h into m/s, correct to one decimal place. 10.09
b A large vat is being filled at a rate of 35 mL/min. Convert this rate to a rate of L/day.

13 Keith and Kent decided to go bushwalking. This travel graph shows their walk. 10.10

a How far did they travel from camp?
b How many stops did they make?
c Find their average speed between their first and second stops.
d How long did it take them to return home?
e Between what times were they walking the fastest?



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